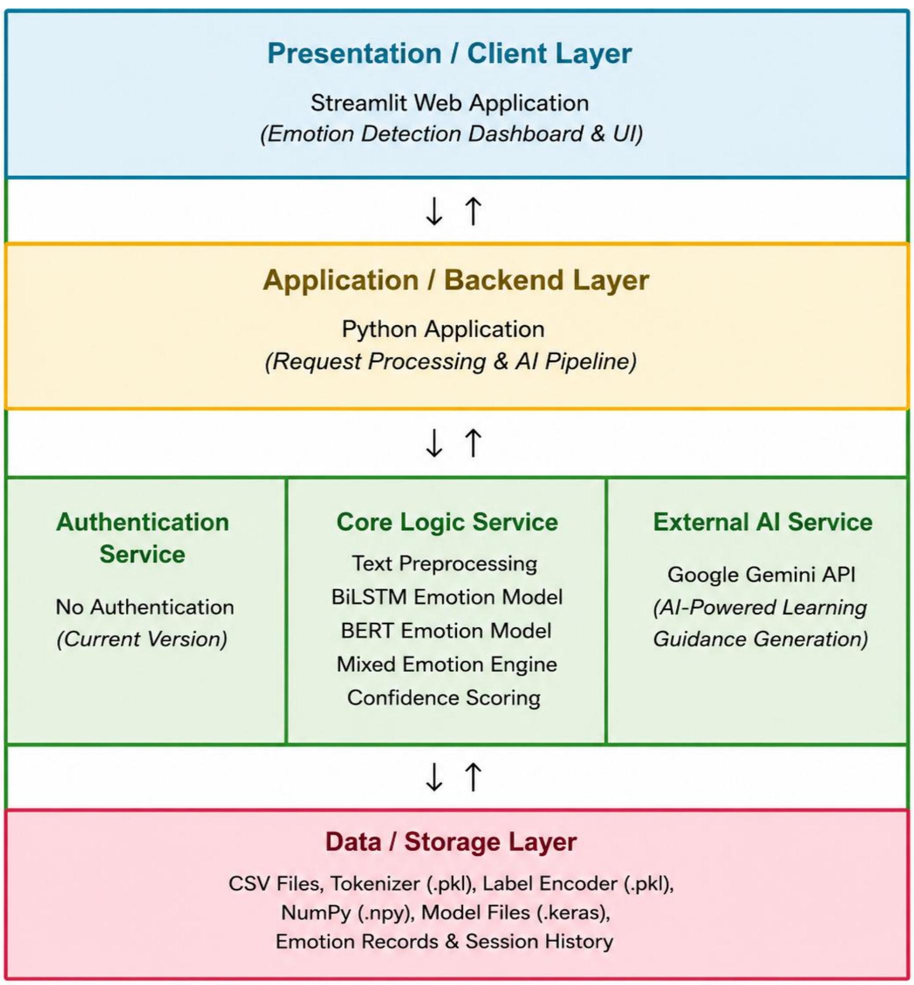
Solution Architecture

Date	28 June 2026
Team ID	xxxxxx
Project Name	Emotion Detection & Learning Support Engine
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides the user interface for entering student text, viewing detected emotions, AI-generated learning guidance, emotion analytics, and dashboards.	Streamlit, HTML, CSS
Application Layer	Handles text preprocessing, emotion prediction, mixed emotion detection, AI response generation, and request processing.	Python
Core Logic Service	Performs text cleaning, tokenization, keyword enhancement, BiLSTM prediction, BERT prediction, confidence scoring, and emotion fusion.	TensorFlow, Hugging Face Transformers, NLTK, Scikit-learn
External AI Integration	Generates personalized and empathetic learning guidance based on detected emotions.	Google Gemini API
Database / Storage	Stores processed datasets, tokenizer, label encoder, trained models, emotion records, session history, and prediction results.	CSV Files, Pickle (.pkl), NumPy (.npy), Keras Models (.keras)
Testing Layer	Validates emotion prediction, AI response generation, preprocessing modules, and application workflow.	PyTest